

Patient Treatment Workflow – Clinical Decision System

This document describes a simplified clinical treatment workflow used in a healthcare system. It includes entities, relationships, diagnostic steps, decision points, and medical rules.

Entities:

- 1 Patient: individual receiving medical care
- 2 Doctor: healthcare professional responsible for diagnosis
- 3 Symptom: observable condition reported by the patient
- 4 Diagnosis: identified medical condition
- 5 Test: medical examination (blood test, imaging, etc.)
- 6 Treatment: medication or procedure administered

Relationships:

- 1 A Patient reports Symptoms
- 2 A Doctor evaluates Symptoms
- 3 Tests support Diagnosis
- 4 Diagnosis determines Treatment
- 5 Treatment is administered to Patient

Process Steps:

- 1 1. Patient reports symptoms
- 2 2. Doctor performs initial evaluation
- 3 3. Order diagnostic tests if necessary
- 4 4. Analyze test results
- 5 5. Confirm diagnosis
- 6 6. Prescribe treatment
- 7 7. Monitor patient response

Decision Points:

- 1 If symptoms are mild → recommend rest and observation
- 2 If test results are inconclusive → order additional tests

- 3 If diagnosis is critical → initiate emergency treatment
- 4 If patient shows adverse reaction → modify treatment

Medical Rules:

- 1 Antibiotics should only be prescribed for confirmed bacterial infections
- 2 Critical patients must be monitored every 2 hours
- 3 All test results must be reviewed before final diagnosis
- 4 Treatment plans must be updated based on patient response

Example Scenario:

A patient reports fever and fatigue. The doctor evaluates symptoms and orders a blood test. If the test confirms infection, antibiotics are prescribed. If the patient responds well, treatment continues; otherwise, the doctor adjusts medication or orders further tests.